CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

Chapter 5: Tales by Dots and Lines

Topic-wise short notes with solved examples

The mean as a balance point, how the mean and median change when data change, the mean and median of data given with frequencies, spreadsheets, and reading and drawing line graphs.

1 Mean and median

- **Mean** (average) $= \dfrac{\text{sum of all the values}}{\text{number of values}}$.
- **Median**: the middle value when the data are arranged in order. With an even number of values, it is the mean of the two middle values.
- Sum of the values = mean $\times$ number of values. This is used again and again.

Example 1.1 Find the mean and median of 8, 3, 10, 13, 4, 6, 7, 7, 8, 8, 5.

SOLUTION

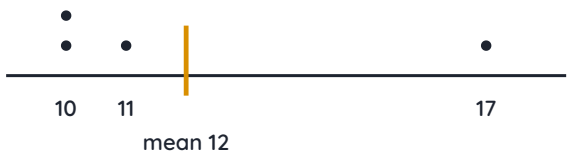
Sum $= 79$ and there are 11 values, so the mean is $\dfrac{79}{11} \approx 7.18$.

In order: 3, 4, 5, 6, 7, 7, 8, 8, 8, 10, 13. The 6th value is 7.

Mean about 7.18; median 7

2 The mean as a balance point

- On a dot plot, the **total distance** of the values below the mean from the mean equals the total distance of the values above it. The mean is the point where the data balance.
- The mean is not always midway between the smallest and largest values.
- There is only one such balance point.



Data 10, 10, 11, 17: distances on the left 2 + 2 + 1 = 5, on the right 5

Example 2.1 Check the balance property for the data 4, 6, 7, 11, 12.

SOLUTION

Mean $= \frac{40}{5} = 8$.

Below the mean: $4 + 2 + 1 = 7$. Above the mean: $3 + 4 = 7$.

The distances balance

3 Adding or removing values

- Including a value **greater than the mean** increases the mean. Including a value **less than the mean** decreases it. Including a value **equal to the mean** leaves it unchanged.
- Removing a value has the opposite effect: removing a value above the mean lowers the mean.
- Several values can be added without changing the mean if their distances above and below the mean balance, for example $m - 3$ and $m + 3$, or $m - 1$, $m - 2$ and $m + 3$.
- The **median** behaves in a similar way: including a value greater than the median cannot lower it, and including a value smaller than the median cannot raise it. The median may also stay the same.

Example 3.1 The mean of 9 numbers is 20. A tenth number, 40, is included. Find the new mean.

SOLUTION

New sum $= 180 + 40 = 220$

22

Example 3.2 The data 3, 5, 8, 8, 9, 12, 14 have median 8. What is the median after 11 is included?

SOLUTION

In order: 3, 5, 8, 8, 9, 11, 12, 14. The two middle values are 8 and 9.

8.5

4 Changing every value

- If the same number k is **added** to every value, the mean increases by k. If k is subtracted, the mean decreases by k.
- If every value is **multiplied** by k, the mean is multiplied by k.
- Reason: $\frac{(x_1 + k) + (x_2 + k) + \ldots + (x_n + k)}{n} = \frac{x_1 + x_2 + \ldots + x_n}{n} + \frac{nk}{n} = a + k.$
- The median changes in exactly the same way.

Example 4.1 The mean height of 24 students, measured with shoes on, is 150.2 cm. The shoes add 1 cm. Find the correct mean height.

SOLUTION

Every value is 1 cm too large, so the mean is 1 cm too large.

149.2 cm

Example 4.2 The mean of a data set is 12. Every value is doubled and then 5 is added. Find the new mean.

SOLUTION

$2 \times 12 + 5$

29

5 Finding an unknown value

- Use: sum $=$ mean $\times$ number of values.

Example 5.1 The mean weight of 10 players is 39.2 kg. Nine of the weights add up to 349 kg. Find the tenth.

SOLUTION

Total $= 39.2 \times 10 = 392$

43 kg

Example 5.2 The mean of 15 values is 25.6. One value was recorded as 3 more than it should be. Find the correct mean.

SOLUTION

Recorded sum $= 25.6 \times 15 = 384$. Correct sum $= 381$.

$\frac{381}{15} = 25.4$

Example 5.3 The mean of 8, 13, 10, 4, 5, 20, y, 10 is 10.375. Find y.

SOLUTION

Sum $= 10.375 \times 8 = 83$, and the known values add up to 70.

$y = 13$

6 Mean and median with frequencies

- When each value occurs several times, multiply each value by its **frequency** before adding.

$$\text{Mean} = \frac{\text{sum of (value} \times \text{frequency)}}{\text{sum of the frequencies}}$$

- Do **not** just average the different values.
- **Median**: add the frequencies one after another (running totals) until you reach the middle position(s).

Family size	3	4	5	6	7	Total
Number of students	4	10	8	5	3	30
Size × number	12	40	40	30	21	143

Example 6.1 Find the mean and median family size from the table above.

SOLUTION

Mean $= \frac{143}{30} \approx 4.77$.

There are 30 values, so the median is the mean of the 15th and 16th values. Running totals: 4, 14, 22. Positions 15 to 22 all hold the value 5.

Mean about 4.77; median 5

7 Spreadsheets

- A **spreadsheet** is a table of **cells** on a computer. Columns are labelled A, B, C, … and rows 1, 2, 3, … The cell in column E and row 5 is **E5**.
- A cell can hold text, a number or a **formula**. A formula begins with =.
- A row or column of cells is written Start:End. B3:G3 means the cells from B3 to G3; D2:D6 means the cells from D2 to D6.

Formula	What it gives
=SUM(B3:G3)	The total of cells B3 to G3
=AVERAGE(B2:B23)	The mean of cells B2 to B23
=MEDIAN(B2:B23)	The median of cells B2 to B23

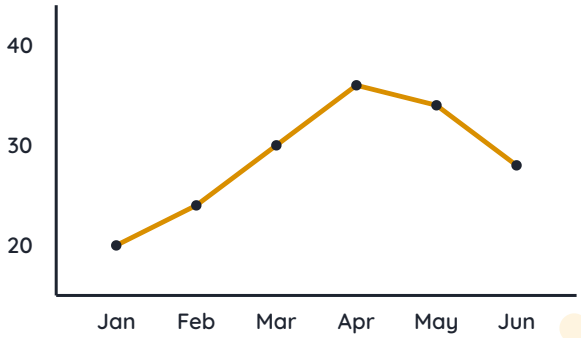
- When a number in the table is changed, every formula that uses it updates automatically.

8 Line graphs

- A **line graph** joins data points by line segments. It is best for showing how a quantity **changes over time**.
- The horizontal axis usually shows time; the vertical axis shows the quantity. Always read the titles, the scale and the key first.
- **Reading a graph in two steps**: (1) identify what is given: axes, scale, units, which line is which; (2) infer and interpret: highest and lowest points, rises and falls, comparisons.
- A **steeper** segment means a faster change. A flat segment means no change.

- Two or more lines on the same axes let us compare trends. With many time points, a line graph is much clearer than a clustered bar graph.
- Ask how the data were collected. This shows what the graph can and cannot tell us.

Month	Jan	Feb	Mar	Apr	May	Jun
Maximum temperature (°C)	20	24	30	36	34	28



Line graph of the monthly maximum temperature (°C)

Example 8.1 From the graph above: (i) in which month is the temperature highest? (ii) Between which two months is the rise greatest? (iii) What is the mean for the six months?

SOLUTION

(i) The highest point is April, at 36.

(ii) The steepest rising segments are Feb to Mar and Mar to Apr, each a rise of 6.

(iii) $\dfrac{20+24+30+36+34+28}{6} = \dfrac{172}{6}$

April; Feb to Mar and Mar to Apr; about 28.7°C

Example 8.2 A line graph shows a shop's visitors over a week: 16, 19, 10, 14, 20, 22, 35 from Monday to Sunday. Describe the trend.

SOLUTION

The lowest point is Wednesday (10). From Wednesday the line rises every day.

Visitors dip mid-week and climb to a peak of 35 on Sunday; the steepest rise is from Saturday to Sunday.

9 Other ways of showing data

- **Infographics** combine pictures, maps, colours and numbers to tell a story quickly. Read the colour scale or key before drawing conclusions.
- An **activity strip** shows how a day is spent: a strip of 48 boxes, one for each half hour, coloured by activity.
- A graph can support only what its data show. "No power cuts" cannot be concluded from a graph of how many homes use electricity for lighting.

10 Common mistakes

- Averaging the distinct values and ignoring their frequencies.
- Finding the median without arranging the data in order.
- Assuming the mean is midway between the smallest and largest values.
- Thinking that adding any new value raises the mean. It depends on whether the value is above or below the mean.
- Reading a line graph without checking the scale on the vertical axis.
- Concluding things about individuals from an average.

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